

Interferometers in silicon photonics

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Index Terms— Silicon Photonics, Mach-Zehnder Interferometer, Michelson Interferometer

I. INTRODUCTION

Interferometers are of significant use on the optical bench. We use the interferometers to accurately measure small differences of optical paths. Scaling down from meters to micrometers, we can build such intricate systems using semiconductor manufacturing technologies. Besides being the proof-of-concept realization, we can use interferometers to design functional photonic chips.

The proposed design will be implemented with a silicon waveguide of height 220 nm and width 500 nm, and with TE polarization. The design includes three parts:

1. Mach-Zehnder interferometers (MZI) with varying length differences. This part is to examine if the predicted signal patterns match the experiment results.
2. Michelson interferometers with loop mirrors. This part is to examine if we can get the halved free spectral range (FSR) as predicted.
3. Simple in/out circuits with varying path lengths to characterize the insertion loss and waveguide performance. This part acts as testing loops.

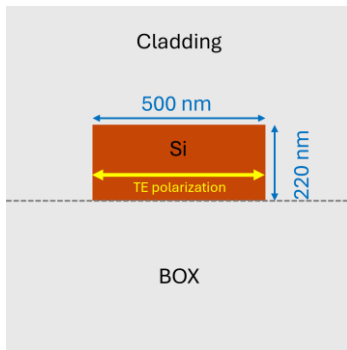


Fig. 1. Geometry of the waveguide (width 500 nm, height 220 nm) and the polarization direction.

II. THEORY

A. Waveguide

A photonic waveguide is composed of a higher index core material and a lower index cladding material. The dimensions and index difference determine the degree of confinement of the optical mode. Normally, the waveguide is designed to be rectangular. For a common SOI process, the height is 220 nm and the width is around 500 nm.

The propagation mode often has a bandwidth, and thus will

experience a dispersion. The information is propagating at a velocity called the group velocity, labeled as n_g .

$$n_g = n - \lambda \frac{dn}{d\lambda} \quad (1)$$

To change the direction of light, we use bent waveguides. Bent structure will introduce bent loss and thus need careful modeling to optimize for the design.

B. Y-Branch splitter

To create different paths, we need splitters. The y-branch splitter has two directions: splitting and combining.

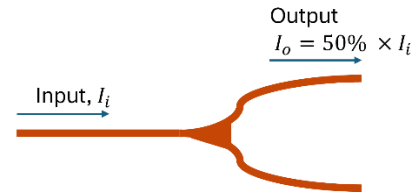


Fig. 2. A 50-50 y-branch splitter.

For an ideal 50-50 y-branch splitter, the output power from one the port is 50% of the input power.

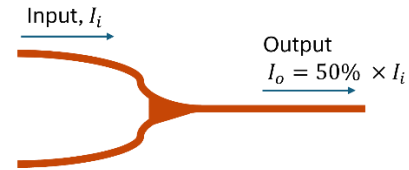


Fig. 3. A 50-50 y-branch combiner.

For a combiner, however, it's important to remember that the input from one port only transmits 50% of the power to the fundamental mode, another 50% to the higher order or radiation mode. The reciprocity of the 50-50 splitting in the case of the combiner thus be port-to-modes, instead of the port-to-ports for the splitter.

We can use the y-branch to split and combine light and build interferometers.

C. Interferometers

Interferometers are devices that combine multiple waves to create interference patterns, from which we can measure quantities like distance, optical index, etc.

Mach-Zehnder interferometers (MZI) constitutes of two beam splitters. The first is to create two lights from one input and the second then combine them into one output.

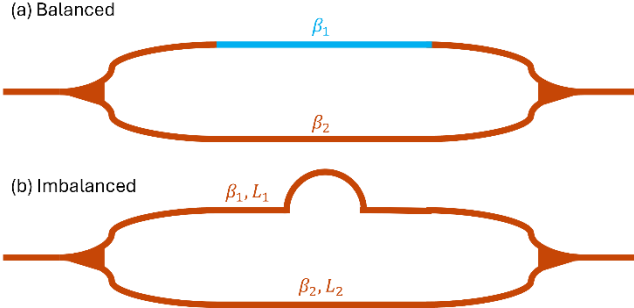


Fig. 4. MZI model. (a) Balanced MZI. (b) Imbalanced MZI.

It's balanced MZI if the two paths are the same, and imbalanced if they are different. The transfer function of MZI is

$$\frac{I_o}{I_i} = \frac{1}{4} \left| e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \right|^2 \quad (2)$$

where $\beta_i = 2\pi n_i / \lambda$ is the propagation constant, n_i is the effective index, α_i is the loss, and L_i is the length of each waveguide.

For lossless waveguides, the transfer function can be written as

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta_1 L_1 - \beta_2 L_2)] \quad (3)$$

If the two waveguide have the same propagation constant but differ in length, as the model shown in Fig 4 (b), then the transfer function is

$$\frac{I_o}{I_i} = \frac{1}{2} \left[1 + \cos\left(\frac{2\pi n}{\lambda} \Delta L\right) \right] \quad (4)$$

where $\Delta L = L_1 - L_2$ is the length difference.

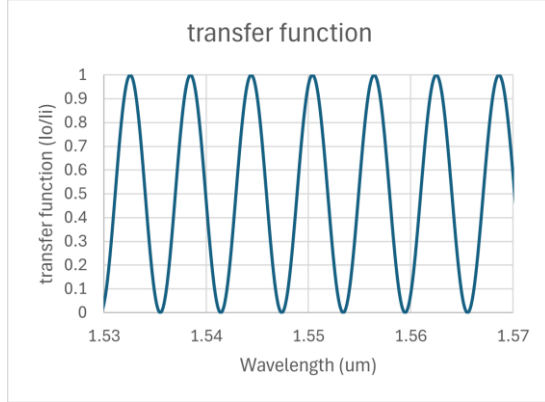


Fig. 5. Plot of the transfer function. Assume the index at 1.5 um is 2.4, with $dn/d\lambda = -1 \mu\text{m}^{-1}$, $\Delta L = 100 \mu\text{m}$.

The transfer function changes sinusoidally with wavelength. The peak-to-peak spacing is call the free spectral range (FSR). Around a particular peak, we can find the FSR by

$$FSR = \frac{\lambda^2}{\Delta L n_g} \quad (5)$$

We can also determine the group index by experimentally measuring FSR.

A Michelson interferometer is composed of one splitter and two mirrors. Since the light will be reflected to the same

modulated waveguide, the accumulated phase difference is twice of the MZI. Hence for the same amount of phase difference, the FSR of Michelson interferometer is half of the MZI's FSR.

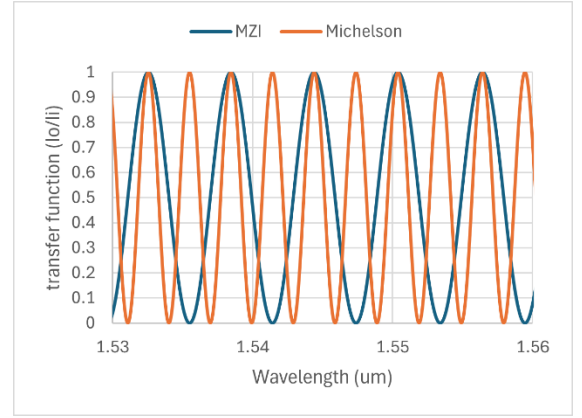


Fig. 6. Plot of the transfer function of MZI and Michelson interferometer. Assume the index at 1.5 um is 2.4, with $dn/d\lambda = -1 \mu\text{m}^{-1}$, $\Delta L = 100 \mu\text{m}$.

III. MODELING AND SIMULATION

A. Waveguide

We use Lumerical MODE solver to simulate the optical mode of the waveguide, and obtain the effective index, group index and the polynomial description of the waveguide.

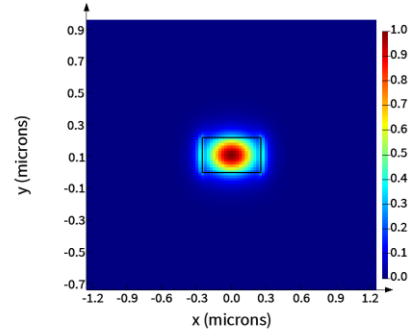


Fig. 7. TE mode profile at 1550 nm simulated using Lumerical MODE. The effective index is 2.44.

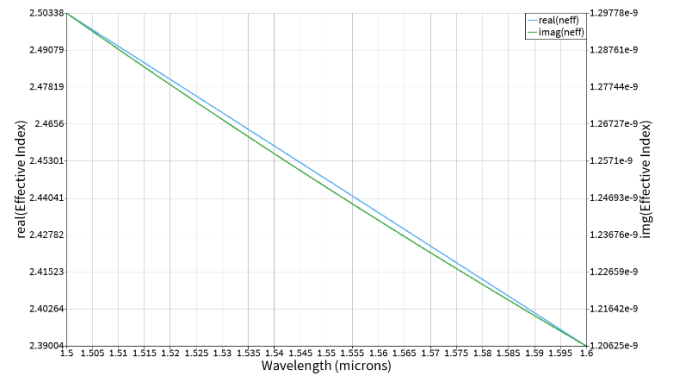


Fig. 8. Effective index dispersion of the TE mode.

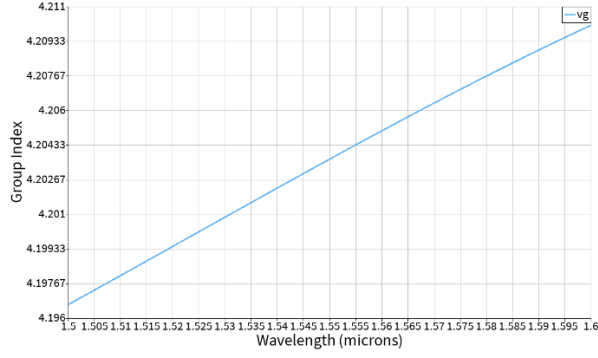


Fig. 9. Group index dispersion of the TE mode.

The polynomial model of the waveguide is

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \quad (6)$$

With the coefficients

$$\begin{aligned} n_1 &= 2.44682 \\ n_2 &= -1.1334 \\ n_3 &= -0.0439845 \\ \lambda_0 &= 1550 \text{ nm} \end{aligned}$$

To characterize the manufactured waveguides, we can design a set of testing loops, with the waveguide length varying from 157 μm to 197 μm in a 20 μm increment.

B. MZI and Michelson Interferometer

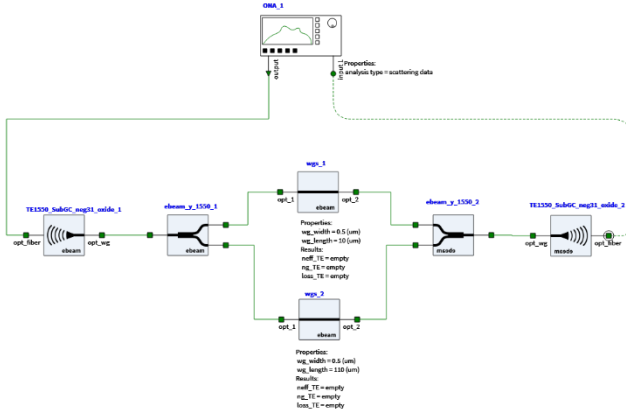


Fig. 10. The MZI model in Lumerical INTERCONNECT.

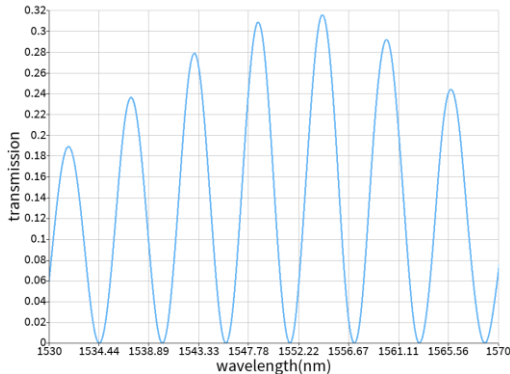


Fig. 11. The transmission spectrum of the MZI model with 100 μm path difference.

TABLE I
DESIGN PARAMETERS OF MZI

ΔL (μm)	Expected FSR (nm)	Simulated FSR (nm)
0	-	-
50	11.5	11.30
100	5.7	5.71
150	3.8	3.81
200	2.9	2.86
250	2.3	2.30

From the simulated result for 50-250 μm ΔL , we can calculate the group index by either 1) curve fitting method 2) direct calculation. The calculated group indices are 4.4 and 4.2, respectively.

TABLE II
DESIGN FOR COMPARE MZI AND MICHELSON
INTERFEROMETER

ΔL (μm)	MZI FSR (nm)	Michelson FSR (nm)
0	-	-
50	11.5	5.7

Use the $\Delta L=50$ μm configuration to compare the MZI and Michelson interferometer.

C. Corner analysis

Since the fabricated waveguides have variations in dimensions, the group index of the waveguides may vary. The ranges of variation are 215.3-233.1 nm in height, 470-510 nm in width, with 220x500 nm as nominal design.

We can perform corner analysis and find the range of the group index variation. From the result shown in table III, the group index may vary between 4.16-4.24, with nominal of 4.18.

TABLE III
CORNER ANALYSIS

Height (nm)	Width (nm)	Group index @ 1550 nm, TE
215.3	470	4.23
215.3	510	4.16
220	500	4.18
233.1	470	4.24
233.1	510	4.17

IV. LAYOUT

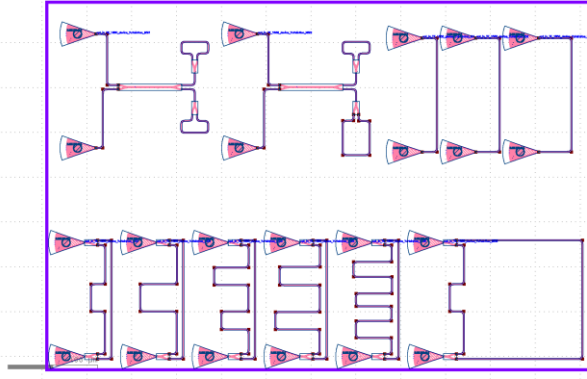


Fig. 12. The layout used for this project.

The lower 6 circuits are MZI with $\Delta L=0, 50, 100, 150, 200, 250$ nm, respectively. The upper left two circuits are Michelson interferometers with $\Delta L=0, 50$ nm. The upper right three are I/O circuits for calibration.

V. FABRICATION AND MEASUREMENT

A. Fabrication

Fabrication is performed at one or more of these: Applied Nanotools and Washington Nanofabrication Facility. The following are the process descriptions.

Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet

etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

Washington Nanofabrication Facility (WNF) silicon photonics process:

The devices were fabricated using 100 keV Electron Beam Lithography [1]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 μm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 °C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 μm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 $\mu\text{C}/\text{cm}^2$ was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using plasma enhanced chemical vapor deposition (PECVD) in an Oxford Plasmalab System 100 with a silane (SiH_4) flow of 13.0 sccm, nitrous oxide (N_2O) flow of 1000.0 sccm, high-purity nitrogen (N_2) flow of 500.0 sccm, pressure at 1400mT, high-frequency RF power of 120W, and a platen temperature of 350C. During deposition, chips rest directly on a silicon carrier wafer and are buffered by silicon pieces on all sides to aid uniformity.

B. Measurement

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

VI. EXPERIMENT RESULTS

This project has 3 different circuit designs: simple I/O, MZI and Michelson interferometers. Since different designs contain different basic components, the basic insertion loss will vary. Fig. 13 shows the comparison between these 3 designs, where the total path lengths for IO, MZI and MSI are 157 μm , 303 μm and 160 μm respectively. Although MSI has one 50-50 splitter and two y splitters more than the simple IO circuit, they have compatible loss. The MZI circuit has slightly lower transmission than the other two, it's most possibly contributed by the length difference.

From Fig. 14 and Fig. 15, we can observe the base line of the transmission for MZI and MSI circuits. While the MSI follows closely with the base line, the MZI spectrum experiences slight distortion when interference happens. Since only one calibration circuit was included in this project, what caused the distortion is yet to decide.

From the transmission spectrum, we extract the FSR at 1550 nm by fitting the spectrum. Table IV lists the result. The deviation of FSR from simulated values are within 1.06%.

We can calculate the by two ways:

- Calculate from FSR for individual result. As shown in table V, the average group index found this way is 4.206 with standard deviation of 0.008.
- Calculate from the curve fitting result. As shown in Fig. 17, the group index found from the slope of the fitted equation is 4.21.

The group index from experiment result lies within the range of 4.16-4.24, and is 0.6% larger than the nominal value of 4.18.

Noted that the although FSR of MSI isn't half of the FSR of MZI of the same ΔL , the value is compatible with the simulated value. Hence the possible reason is the design didn't create the mirror reflection, instead, the lights just travel around the closed loop. In this way, the path difference remains the same and the FSR of MSI circuit is close to it's counterpart of MZI circuit.

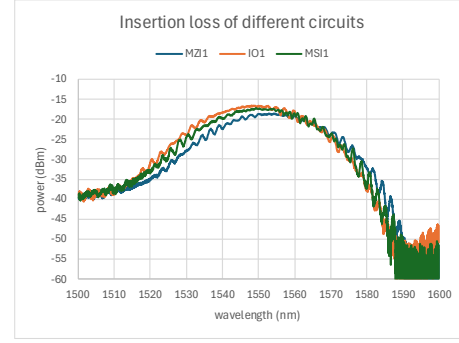


Fig. 13. Comparison of the insertion loss of IO, MSI and MZI circuit designs.

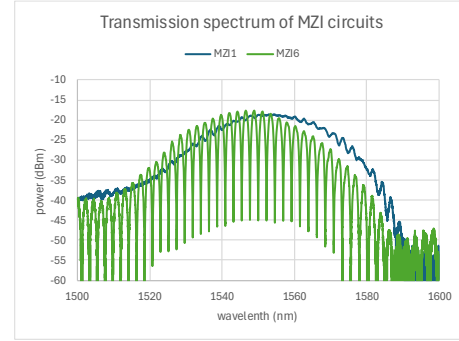


Fig. 14. The transmission spectrum of MZI circuits, where the MZI1 with $\Delta L = 0 \mu\text{m}$ and MZI6 with $\Delta L = 250 \mu\text{m}$.

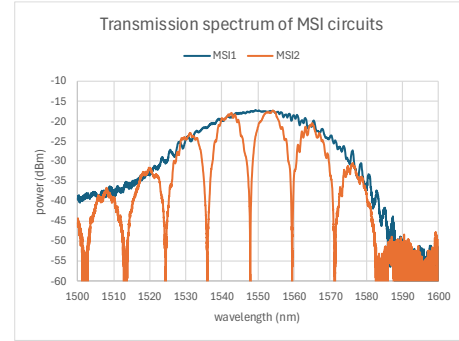


Fig. 15. The transmission spectrum of MSI circuits, where the MSI1 with $\Delta L = 0 \mu\text{m}$ and MSI2 with $\Delta L = 50 \mu\text{m}$.

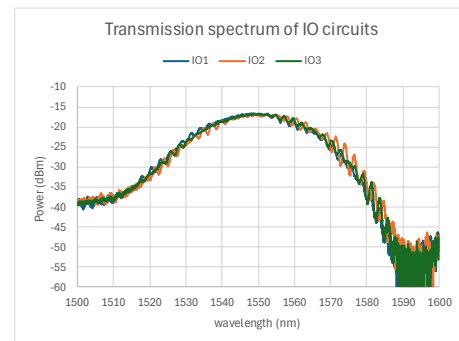


Fig. 16. The transmission spectrum of IO circuits.

TABLE IV
FSR RESULTS

Type	ΔL (μm)	Simulated FSR (nm)	Experimental FSR (nm)	Difference (%)
MZI	50	11.30	11.42	1.06
	100	5.71	5.69	-0.35
	150	3.81	3.81	0.00
	200	2.86	2.86	0.00
	250	2.30	2.29	-0.43
MSI	50	11.54	11.62	0.69

TABLE V
GROUP INDEX FROM MZI CIRCUITS

ΔL (μm)	Group Index
50	4.21
100	4.22
150	4.20
200	4.20
250	4.20
Average	4.206
Standard Deviation	0.008

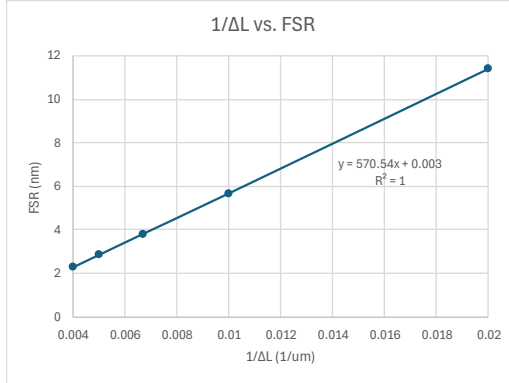


Fig. 17. Curve fitting result for finding group index. The calculated group index is 4.21.

V. CONCLUSION

We designed 3 I/O circuits, 6 MZI circuits and 2 MSI circuits in this project. The experimental FSRs match the simulated values well, with deviation within 1.06%. The group index found from MZI circuits is 4.206 in average, with standard deviation of 0.008. This lies within the range calculated from corner analysis, indicating the fabrication variation has been well defined.

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